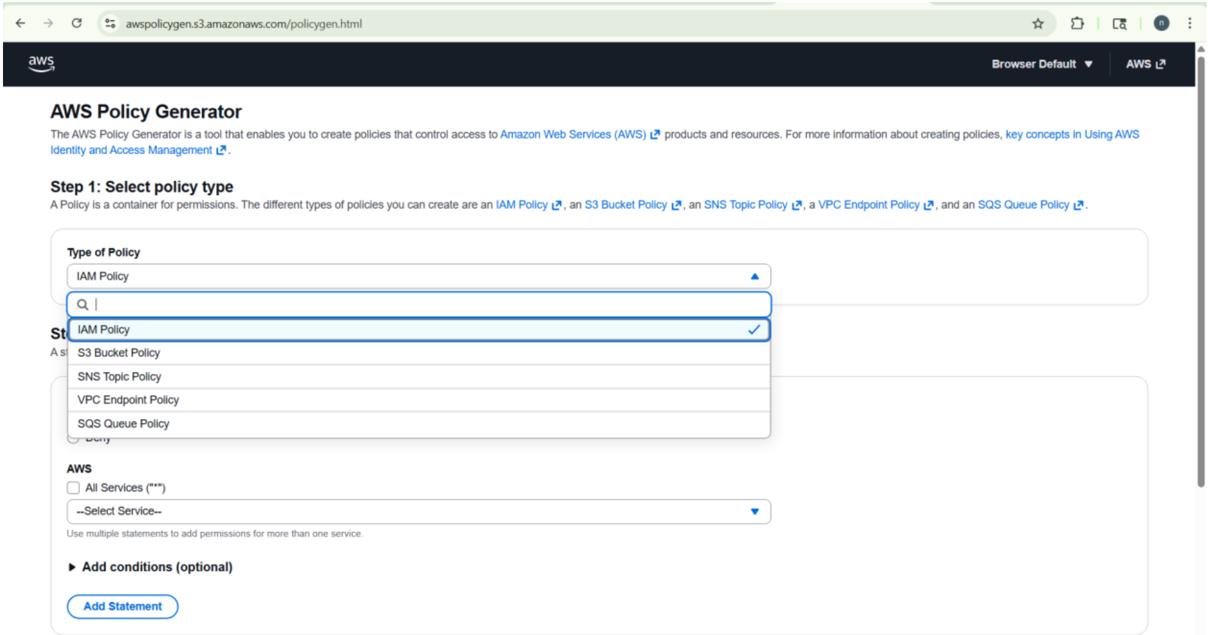
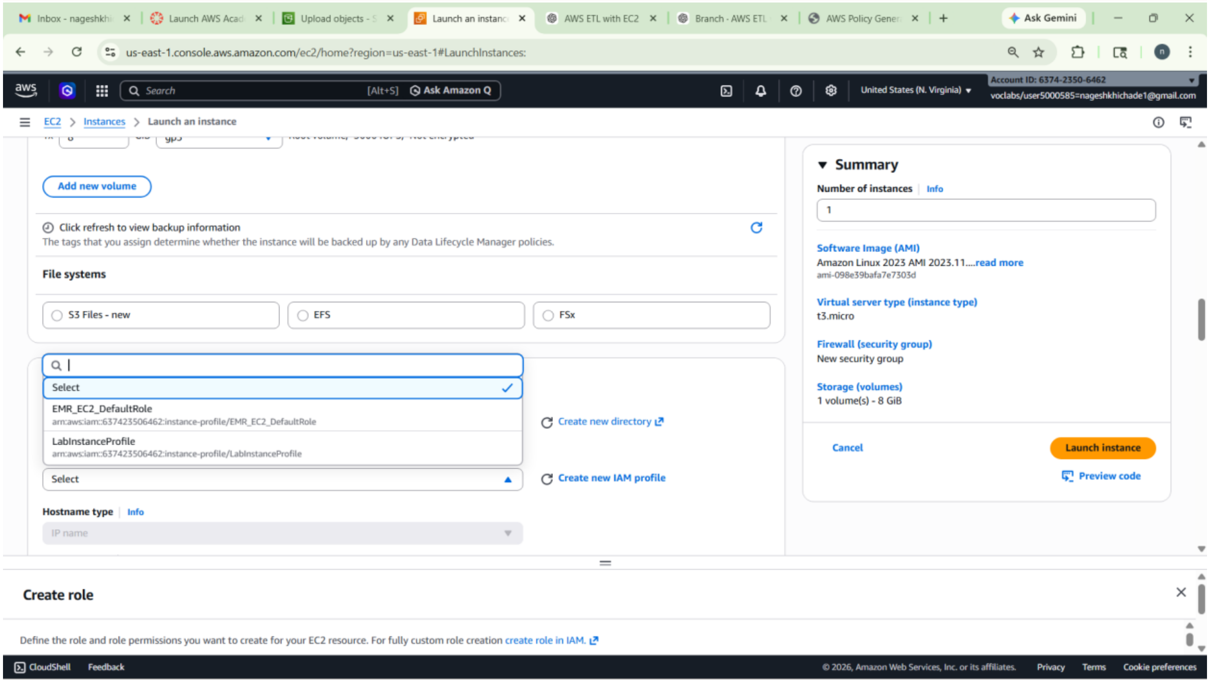

awspolicygen.s3.amazonaws.com/policygen.html
aws
Browser Default
AWS
AWS Policy Generator
The AWS Policy Generator is a tool that enables you to create policies that control access to Amazon Web Services (AWS) products and resources. For more information about creating policies, key concepts in Using AWS Identity and Access Management.
Step 1: Select policy type
A Policy is a container for permissions. The different types of policies you can create are an IAM Policy, an S3 Bucket Policy, an SNS Topic Policy, a VPC Endpoint Policy, and an SQS Queue Policy.
Type of Policy
IAM Policy
IAM Policy
S3 Bucket Policy
SNS Topic Policy
VPC Endpoint Policy
SQS Queue Policy
AWS
All Services ('*')
--Select Service--
Use multiple statements to add permissions for more than one service.
Add conditions (optional)
Add Statement


Inbox - nageshkh
Launch AWS Acad
Upload objects - S
Launch an instanc
AWS ETL with EC2
Branch · AWS ETL
AWS Policy Gener
Ask Gemini
us-east-1.console.aws.amazon.com/ec2/home?region=us-east-1#LaunchInstances:
Search
[Alt+S]
Ask Amazon Q
United States (N. Virginia)
Account ID: 6374-2350-6462
voclabs/user5000585=nageshkhichade1@gmail.com
EC2
Instances
Launch an instance
Add new volume
Click refresh to view backup information
The tags that you assign determine whether the instance will be backed up by any Data Lifecycle Manager policies.
File systems
S3 Files - new
EFS
FSx
Select
EMR_EC2_DefaultRole
arn:aws:iam::637423506462:instance-profile/EMR_EC2_DefaultRole
LabInstanceProfile
arn:aws:iam::637423506462:instance-profile/LabInstanceProfile
Create new directory
Select
Create new IAM profile
Hostname type
Info
IP name
Summary
Number of instances
Info
1
Software Image (AMI)
Amazon Linux 2023 AMI 2023.11....read more
ami-098e39bafa7e7303d
Virtual server type (instance type)
t3.micro
Firewall (security group)
New security group
Storage (volumes)
1 volume(s) - 8 GiB
Cancel
Launch instance
Preview code
Create role
Define the role and role permissions you want to create for your EC2 resource. For fully custom role creation create role in IAM.
CloudShell
Feedback
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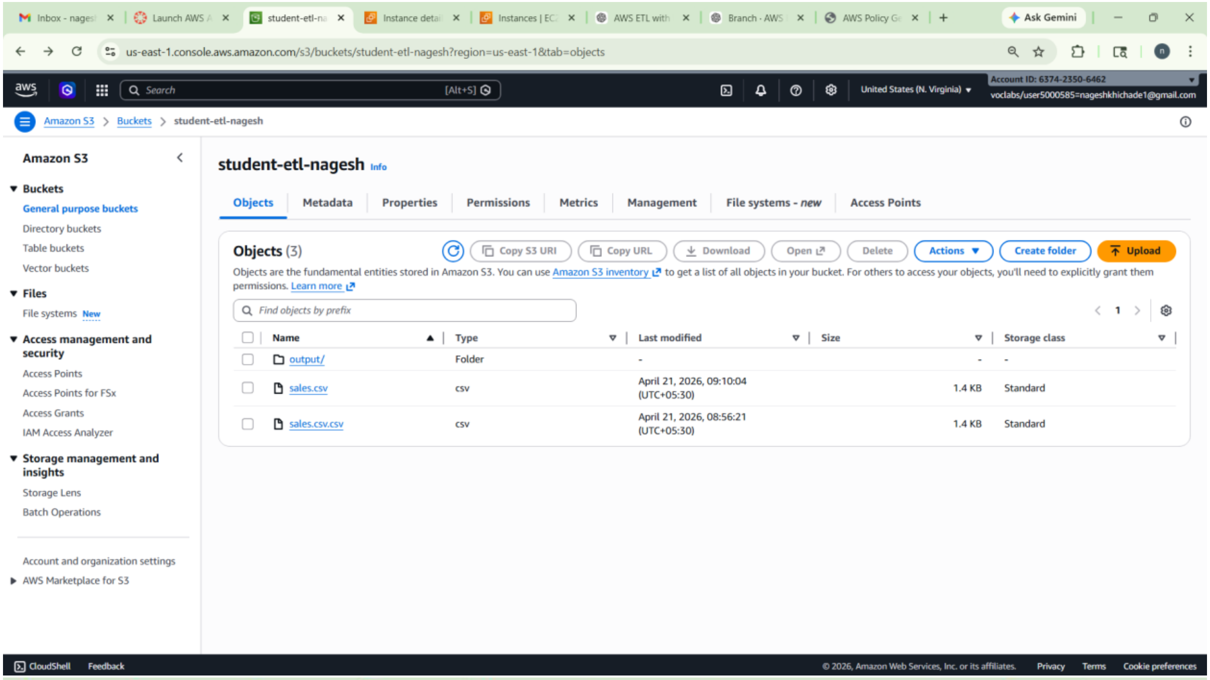
Privacy
Terms
Cookie preferences


us-east-1.console.aws.amazon.com/s3/buckets/student-etl-nagesh?region=us-east-1&tab=objects
Account ID: 6374-2350-6462
voclabs/user5000585=nageshkhichade1@gmail.com
United States (N. Virginia)
Amazon S3 > Buckets > student-etl-nagesh
Amazon S3
Buckets
General purpose buckets
Directory buckets
Table buckets
Vector buckets
Files
File systems New
Access management and security
Access Points
Access Points for FSx
Access Grants
IAM Access Analyzer
Storage management and insights
Storage Lens
Batch Operations
Account and organization settings
AWS Marketplace for S3
student-etl-nagesh Info
Objects
Metadata
Properties
Permissions
Metrics
Management
File systems - new
Access Points
Objects (3)
Copy S3 URI
Copy URL
Download
Open
Delete
Actions
Create folder
Upload
Objects are the fundamental entities stored in Amazon S3. You can use Amazon S3 inventory to get a list of all objects in your bucket. For others to access your objects, you'll need to explicitly grant them permissions. Learn more
Find objects by prefix
Name
Type
Last modified
Size
Storage class
output/
Folder
sales.csv
csv
April 21, 2026, 09:10:04 (UTC+05:30)
1.4 KB
Standard
sales.csv.csv
csv
April 21, 2026, 08:56:21 (UTC+05:30)
1.4 KB
Standard
CloudShell
Feedback
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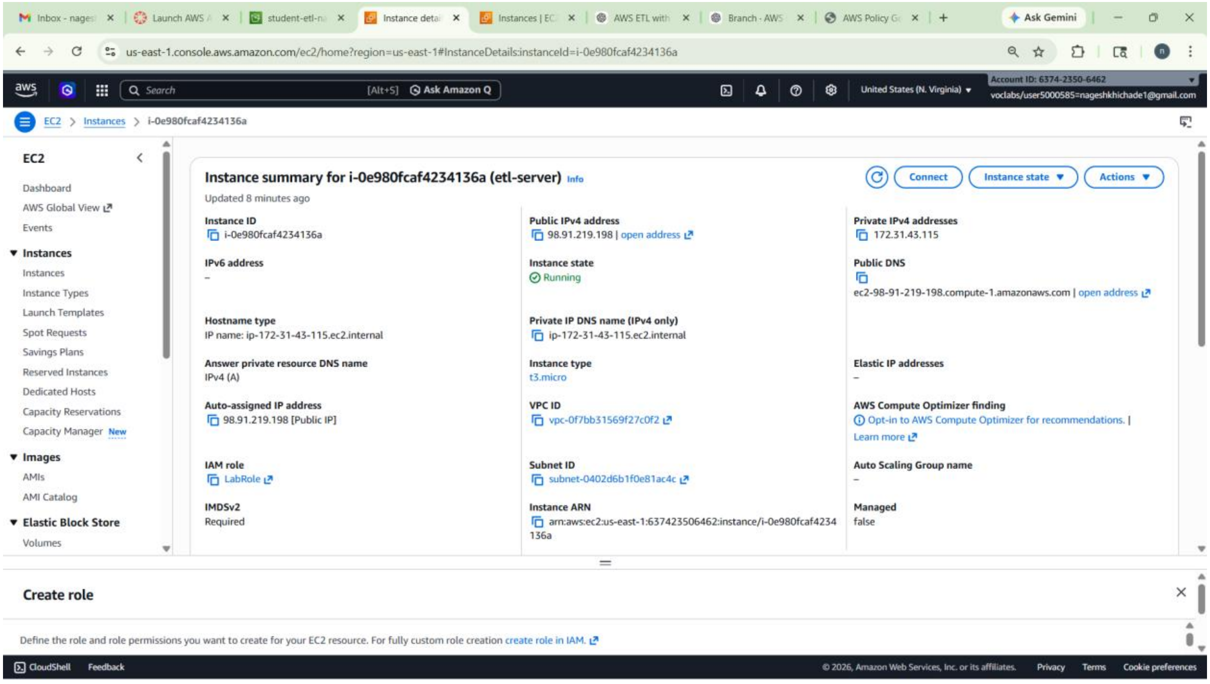
Terms
Cookie preferences


us-east-1.console.aws.amazon.com/ec2/home?region=us-east-1#InstanceDetails:instanceId=i-0e980fcaf4234136a
Account ID: 6374-2350-6462
voclabs/user5000585=nageshkhichade1@gmail.com
EC2 > Instances > i-0e980fcaf4234136a
EC2
Dashboard
AWS Global View
Events
Instances
Instances
Instance Types
Launch Templates
Spot Requests
Savings Plans
Reserved Instances
Dedicated Hosts
Capacity Reservations
Capacity Manager New
Images
AMIs
AMI Catalog
Elastic Block Store
Volumes
Instance summary for i-0e980fcaf4234136a (etl-server) Info
Updated 8 minutes ago
Connect
Instance state
Actions
Instance ID
i-0e980fcaf4234136a
Public IPv4 address
98.91.219.198 | open address
Private IPv4 addresses
172.31.43.115
IPv6 address
–
Instance state
Running
Public DNS
ec2-98-91-219-198.compute-1.amazonaws.com | open address
Hostname type
IP name: ip-172-31-43-115.ec2.internal
Private IP DNS name (IPv4 only)
ip-172-31-43-115.ec2.internal
Answer private resource DNS name
IPv4 (A)
Instance type
t3.micro
Elastic IP addresses
–
Auto-assigned IP address
98.91.219.198 [Public IP]
VPC ID
vpc-0f7bb31569f27c0f2
AWS Compute Optimizer finding
Opt-in to AWS Compute Optimizer for recommendations. | Learn more
IAM role
LabRole
Subnet ID
subnet-0402d6b1f0e81ac4c
Auto Scaling Group name
–
IMDSv2
Required
Instance ARN
arn:aws:ec2:us-east-1:637423506462:instance/i-0e980fcaf4234136a
Managed
false
Create role
Define the role and role permissions you want to create for your EC2 resource. For fully custom role creation create role in IAM.
CloudShell
Feedback
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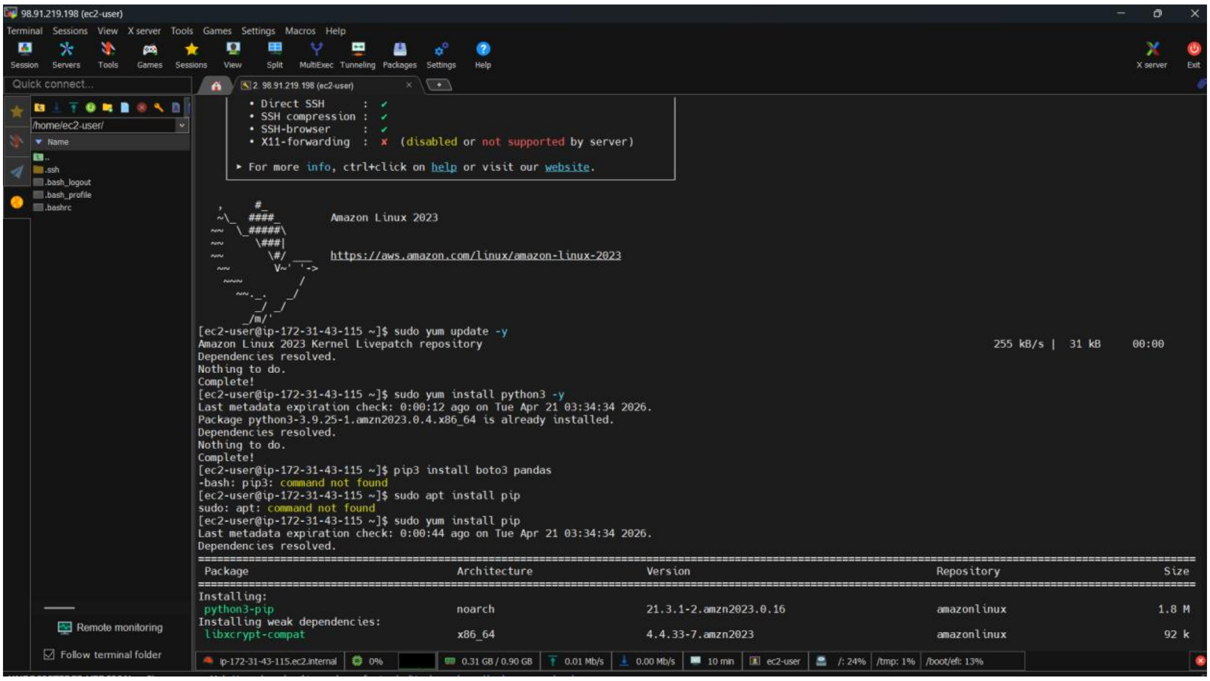
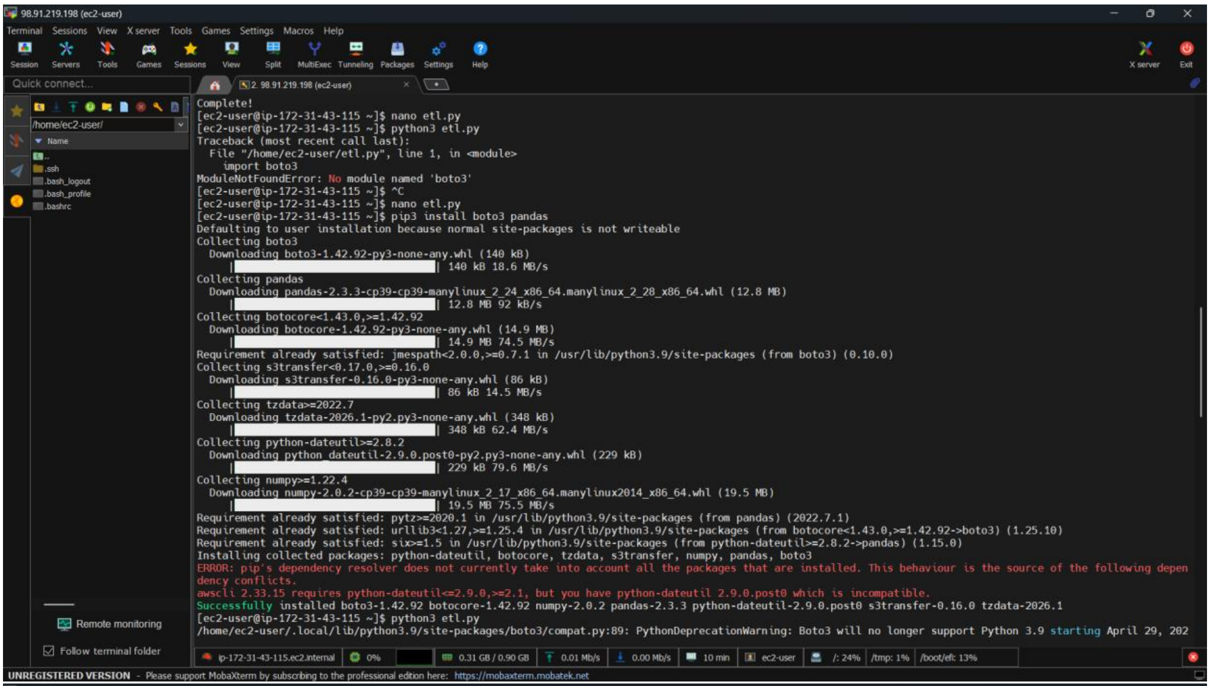
Privacy
Terms
Cookie preferences



Python Script:

```
import boto3

import pandas as pd


bucket_name = "student-etl-nagesh"

input_file = "sales.csv"

output_file = "output/filtered_sales.csv"


s3 = boto3.client('s3')


# Download file from S3

s3.download_file(bucket_name, input_file, '/tmp/sales.csv')


print("Downloaded file from S3")


# Read CSV

df = pd.read_csv('/tmp/sales.csv')
```

```
print("Original Data:")
print(df.head())

# Filter records
filtered_df = df[df['amount'] > 1000]

print("Filtered Data:")
print(filtered_df)

# Save filtered file
filtered_df.to_csv('/tmp/filtered_sales.csv', index=False)

# Upload back to S3
s3.upload_file('/tmp/filtered_sales.csv', bucket_name, output_file)

print("Uploaded filtered file to S3 → output/filtered_sales.csv")
```


98.91.219.198 (ec2-user)
2. 98.91.219.198 (ec2-user)
Complete!
[ec2-user@ip-172-31-43-115 ~]$ nano etl.py
[ec2-user@ip-172-31-43-115 ~]$ python3 etl.py
Traceback (most recent call last):
File "/home/ec2-user/etl.py", line 1, in <module>
import boto3
ModuleNotFoundError: No module named 'boto3'
[ec2-user@ip-172-31-43-115 ~]$ ^C
[ec2-user@ip-172-31-43-115 ~]$ nano etl.py
[ec2-user@ip-172-31-43-115 ~]$ pip3 install boto3 pandas
Defaulting to user installation because normal site-packages is not writeable
Collecting boto3
Downloading boto3-1.42.92-py3-none-any.whl (140 kB)
140 kB 18.6 MB/s
Collecting pandas
Downloading pandas-2.3.3-cp39-cp39-manylinux_2_24_x86_64.manylinux_2_28_x86_64.whl (12.8 MB)
12.8 MB 92 kB/s
Collecting botocore<1.43.0,>=1.42.92
Downloading botocore-1.42.92-py3-none-any.whl (14.9 MB)
14.9 MB 74.5 MB/s
Requirement already satisfied: jmespath<2.0.0,>=0.7.1 in /usr/lib/python3.9/site-packages (from boto3) (0.10.0)
Collecting s3transfer<0.17.0,>=0.16.0
Downloading s3transfer-0.16.0-py3-none-any.whl (86 kB)
86 kB 14.5 MB/s
Collecting tzdata>=2022.7
Downloading tzdata-2026.1-py2.py3-none-any.whl (348 kB)
348 kB 62.4 MB/s
Collecting python-dateutil>=2.8.2
Downloading python_dateutil-2.9.0.post0-py2.py3-none-any.whl (229 kB)
229 kB 79.6 MB/s
Collecting numpy>=1.22.4
Downloading numpy-2.0.2-cp39-cp39-manylinux_2_17_x86_64.manylinux2014_x86_64.whl (19.5 MB)
19.5 MB 75.5 MB/s
Requirement already satisfied: pytz>=2020.1 in /usr/lib/python3.9/site-packages (from pandas) (2022.7.1)
Requirement already satisfied: urllib3<1.27,>=1.25.4 in /usr/lib/python3.9/site-packages (from botocore<1.43.0,>=1.42.92->boto3) (1.25.10)
Requirement already satisfied: six>=1.5 in /usr/lib/python3.9/site-packages (from python-dateutil>=2.8.2->pandas) (1.15.0)
Installing collected packages: python-dateutil, botocore, tzdata, s3transfer, numpy, pandas, boto3
ERROR: pip's dependency resolver does not currently take into account all the packages that are installed. This behaviour is the source of the following dependency conflicts.
awscli 2.33.15 requires python-dateutil<=2.9.0,>=2.1, but you have python-dateutil 2.9.0.post0 which is incompatible.
Successfully installed boto3-1.42.92 botocore-1.42.92 numpy-2.0.2 pandas-2.3.3 python-dateutil-2.9.0.post0 s3transfer-0.16.0 tzdata-2026.1
[ec2-user@ip-172-31-43-115 ~]$ python3 etl.py
/home/ec2-user/.local/lib/python3.9/site-packages/boto3/compat.py:89: PythonDeprecationWarning: Boto3 will no longer support Python 3.9 starting April 29, 202
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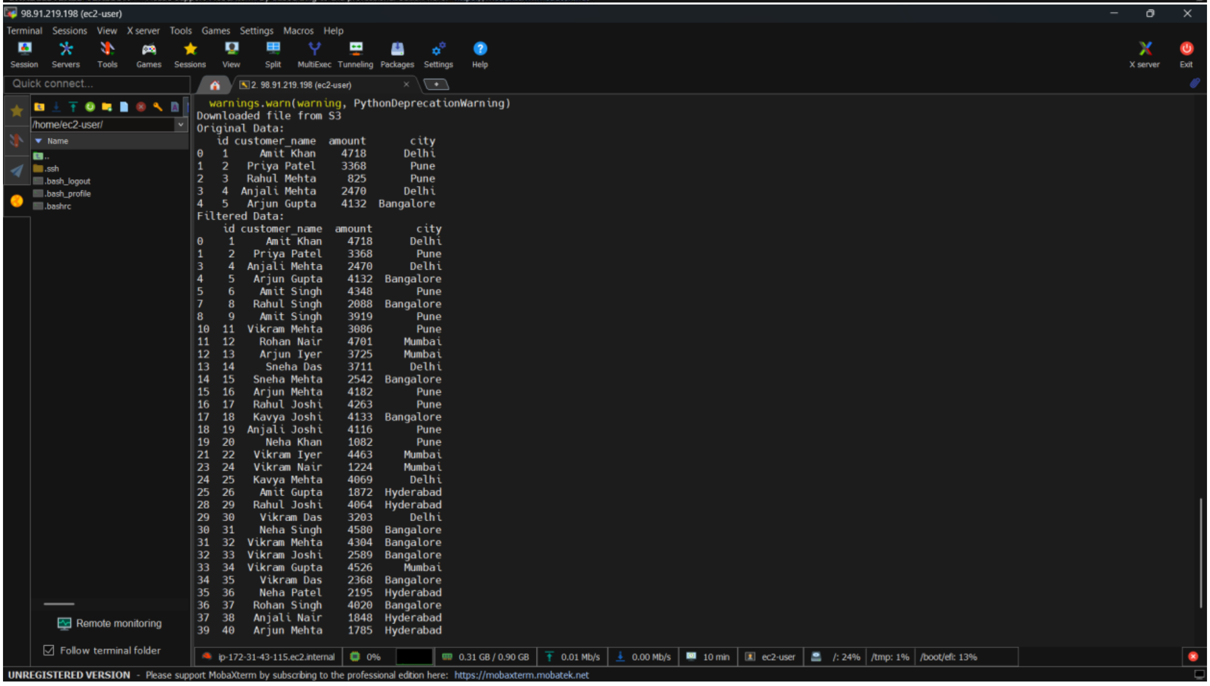
98.91.219.198 (ec2-user)
2. 98.91.219.198 (ec2-user)
warnings.warn(warning, PythonDeprecationWarning)
Downloaded file from S3
Original Data:
id customer_name amount city
0 1 Amit Khan 4718 Delhi
1 2 Priya Patel 3368 Pune
2 3 Rahul Mehta 825 Pune
3 4 Anjali Mehta 2470 Delhi
4 5 Arjun Gupta 4132 Bangalore
Filtered Data:
id customer_name amount city
0 1 Amit Khan 4718 Delhi
1 2 Priya Patel 3368 Pune
3 4 Anjali Mehta 2470 Delhi
4 5 Arjun Gupta 4132 Bangalore
5 6 Amit Singh 4348 Pune
7 8 Rahul Singh 2088 Bangalore
8 9 Amit Singh 3919 Pune
10 11 Vikram Mehta 3086 Pune
11 12 Rohan Nair 4701 Mumbai
12 13 Arjun Iyer 3725 Mumbai
13 14 Sneha Das 3711 Delhi
14 15 Sneha Mehta 2542 Bangalore
15 16 Arjun Mehta 4182 Pune
16 17 Rahul Joshi 4263 Pune
17 18 Kavya Joshi 4133 Bangalore
18 19 Anjali Joshi 4116 Pune
19 20 Neha Khan 1082 Pune
21 22 Vikram Iyer 4463 Mumbai
23 24 Vikram Nair 1224 Mumbai
24 25 Kavya Mehta 4069 Delhi
25 26 Amit Gupta 1872 Hyderabad
28 29 Rahul Joshi 4064 Hyderabad
29 30 Vikram Das 3203 Delhi
30 31 Neha Singh 4580 Bangalore
31 32 Vikram Mehta 4304 Bangalore
32 33 Vikram Joshi 2589 Bangalore
33 34 Vikram Gupta 4526 Mumbai
34 35 Vikram Das 2368 Bangalore
35 36 Neha Patel 2195 Hyderabad
36 37 Rohan Singh 4020 Bangalore
37 38 Anjali Nair 1848 Hyderabad
39 40 Arjun Mehta 1785 Hyderabad
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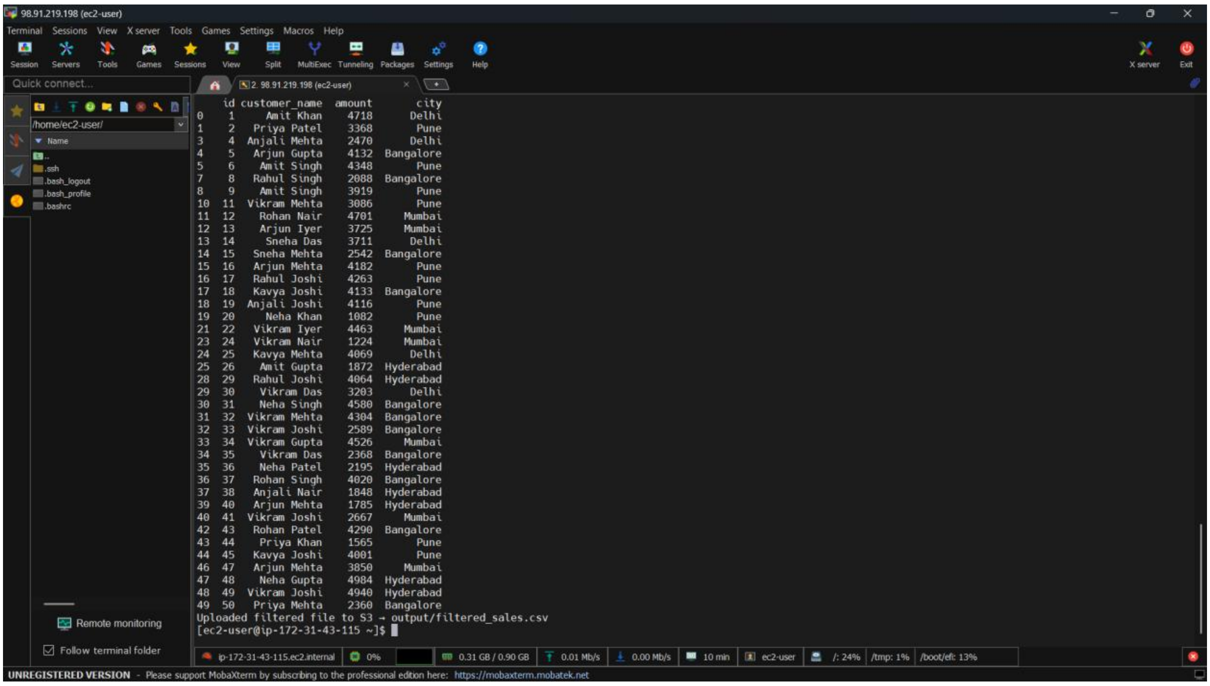

```
    id customer_name  amount       city
0    1     Amit Khan    4718      Delhi
1    2   Priya Patel    3368       Pune
3    4  Anjali Mehta    2470      Delhi
4    5   Arjun Gupta    4132  Bangalore
5    6    Amit Singh    4348       Pune
7    8   Rahul Singh    2088  Bangalore
8    9    Amit Singh    3919       Pune
10  11  Vikram Mehta    3086       Pune
11  12    Rohan Nair    4701     Mumbai
12  13    Arjun Iyer    3725     Mumbai
13  14     Sneha Das    3711      Delhi
14  15   Sneha Mehta    2542  Bangalore
15  16   Arjun Mehta    4182       Pune
16  17   Rahul Joshi    4263       Pune
17  18   Kavya Joshi    4133  Bangalore
18  19  Anjali Joshi    4116       Pune
19  20     Neha Khan    1082       Pune
21  22   Vikram Iyer    4463     Mumbai
23  24   Vikram Nair    1224     Mumbai
24  25   Kavya Mehta    4069      Delhi
25  26    Amit Gupta    1872  Hyderabad
28  29   Rahul Joshi    4064  Hyderabad
29  30    Vikram Das    3203      Delhi
30  31    Neha Singh    4580  Bangalore
31  32  Vikram Mehta    4304  Bangalore
32  33  Vikram Joshi    2589  Bangalore
33  34  Vikram Gupta    4526     Mumbai
34  35    Vikram Das    2368  Bangalore
35  36    Neha Patel    2195  Hyderabad
36  37   Rohan Singh    4020  Bangalore
37  38   Anjali Nair    1848  Hyderabad
39  40   Arjun Mehta    1785  Hyderabad
40  41  Vikram Joshi    2667     Mumbai
42  43   Rohan Patel    4290  Bangalore
43  44    Priya Khan    1565       Pune
44  45   Kavya Joshi    4001       Pune
46  47   Arjun Mehta    3850     Mumbai
47  48    Neha Gupta    4984  Hyderabad
48  49  Vikram Joshi    4940  Hyderabad
49  50   Priya Mehta    2360  Bangalore
Uploaded filtered file to S3 → output/filtered_sales.csv
[ec2-user@ip-172-31-43-115 ~]$
```